Technical Framework for Village Needs, Dependency Routes, and Governance in Nepal

Uddhav P. Gautam

Purpose and Scope

This document defines a technical framework for understanding village needs in Nepal across the full continuum from fully remote settlements to fully connected villages. Its first objective is to identify and organize local needs as a rich model rather than as an unstructured list. Each need can be classified, categorized, and extended with metadata such as scope, dependencies, barriers, feasibility, and evidence. The framework then maps those needs onto a governance tree that runs from household leaf nodes upward through neighbor groups, wards, villages, and, where required, higher levels of government.

Within this model, governance is treated as the process by which needs are satisfied, coordinated, escalated, and tracked across levels. The core analytical concept is the dependency route: the path from a household to the node that satisfies a given need. Shorter routes imply less coordination overhead, fewer points of failure, and greater household self-reliance. Longer routes imply greater institutional dependence and fragility. This makes route minimization a design principle for resilient local governance. In this document, e-governance is not the starting point; it is the operational layer through which identified needs are processed and managed.

The long-term purpose of this document is implementation. It is intended to serve as a specification and guideline base for software that models village needs, represents governance structure, analyzes dependency routes, and supports planning for villages in Nepal.

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1 Introduction

This document aims to identify the full range of local village needs in Nepal and to model them as a structured system. The central task is not merely to list needs, but to make each need analytically useful by classifying it, categorizing it, and enriching it with metadata. In this framework, a need can carry information about the scope at which it can be satisfied, the dependencies it has on other needs, the barriers that keep it at a higher level, and the evidence or planning judgment that supports its placement.

Nepal's topography—from the high Himalayas to the Terai plains—and its uneven infrastructure produce villages under radically different conditions. Some settlements are **fully remote**: there are no motor roads, no grid electricity, and access may be only by foot or mule. Other settlements are **fully connected**: they have roads, electricity, mobile service, and internet access. Most villages lie somewhere between these two extremes. A useful framework must therefore describe the full continuum rather than assume one standard village form.

This document models this reality through a governance tree. Households are treated as the leaf nodes of the tree, and higher levels such as neighbor groups, wards, villages, and larger government institutions form the upper layers. Once local needs are identified and richly described, they can be mapped onto this tree to show where they are satisfied and how they are processed through governance structures. In that sense, e-governance is the operational mechanism for processing needs through the system, but the foundation of the work is the structured modeling of the needs themselves.

1.1 Design Goal: Minimize Dependency Routes

The primary goal of this system is **village self-sufficiency**. Every need a household has creates a *dependency route*—a path from the household to whichever node satisfies that need. The length of that route is measured in scope hops from H to N to W to V. That length is a direct measure of how much external coordination, governance, and fragility the household accepts.

Route length 0:
Household satisfies the need itself. No dependency. Maximum resilience. Example: own rice production, own firewood.

Route length 1:
Household depends on its neighbor group. Coordination remains minimal because the group usually contains 2–5 families. Example: labor exchange through *parma* and shared tools.

Route length 2:
Household depends on its ward. Requires ward-level infrastructure and governance. Example: piped water, ward trail.

Route length 3:
Household depends on the village. Requires village-wide institutions. Example: school, health post, haat bazaar.

These dependency routes are analogous to **layers in a software framework** or **layers of government**. Each hop adds a layer of coordination, a potential point of failure, and a governance cost. The insight:

> *The shorter the dependency routes, the simpler the system, the easier it is to govern, and the more self-sufficient the village becomes. The goal is to push every need to the lowest possible scope level.*

Irreducible dependencies. Some needs have a minimum scope that cannot be reduced. A school requires enough children and therefore usually sits at ward or village level. A haat bazaar

requires enough traders and therefore sits at village level. A hospital requires specialist infrastructure and remains an external dependency. These set a floor on route length. The optimization target is simple: for every need where the scope *can* be lower, push it down.

Self-sufficiency score. The system measures this through two metrics described in Part X:

- **Average route length:** The mean number of scope hops across all household dependencies. Lower is better.
- **Push-down opportunities:** Needs currently satisfied at a higher scope than their minimum possible scope. These are actionable targets for increasing self-sufficiency.

Document structure: Part I covers survival basics and includes water, energy, food, shelter, and clothing. Part II covers transport and communication. Part III covers services and includes education and health. Part IV covers governance and economy. Part V covers environment, conservation, and sanitation. Part VI covers social life and livelihood. Part VII covers infrastructure, security, and governance. Part VIII covers the fully independent village. Part IX covers the technical specification. Part X covers decision and feasibility review. Part XI contains references.

Part I

Part I: Survival Basics

2 Water

2.1 Drinking & Domestic

- Springs such as *kuwa* and *dhara*, along with wells, rivers, and streams
- Rainwater harvesting, rooftop collection
- Piped supply, gravity-fed systems
- Water purification: boiling, filters, chlorine, UV
- Storage: tanks, *ghagri*, *gagri*
- Traditional sources include *ghaite* stone spouts and *hitri*
- Water user committees, maintenance
- Seasonal scarcity: dry season, monsoon turbidity
- Water quality testing

2.2 Irrigation

- Canals called *kulo* and diversion weirs
- Lift irrigation, pump irrigation
- Drip and sprinkler systems where they are feasible
- Traditional *pani ghatta* water mills
- Pond, reservoir
- Snowmelt and glacier-fed sources in high-altitude areas

2.3 Irrigation Equipment & Machinery

To make irrigation easier—especially where gravity-fed canals are insufficient or land needs preparation—villages need at least the following.

Water lifting & conveyance:

- **Pump:** Diesel, electric, or solar—to lift water from well, river, stream, pond when source is below field
- **Pipes & hose:** PVC, HDPE, flexible hose—to convey water from source to field; suction and delivery pipes
- **Hand/treadle pump:** Where no fuel or electricity—human-powered lift

Land preparation & earthwork:

- **Power tiller:** A two-wheel tractor used for plowing, tilling, and land leveling. It helps field leveling for even water distribution and reduces runoff. It is common in the Terai and in accessible hill areas.
- **Tractor:** A four-wheel machine used where terrain and roads allow. It supports plowing, tilling, pulling trailers, and sometimes pumping. It is suited to heavier work.
- **Attachments:** Plow, cultivator, leveler, ridger—for tiller/tractor
- **Shovel, spade, *kodalo*:** Manual canal digging, repair—always needed; bare minimum when no machinery

Canal & distribution:

- **Canal lining:** Cement, plastic sheet—reduces seepage, saves water
- **Gates, valves:** To control flow, allocate water
- **Drip kit:** Main line, laterals, emitters—efficient for vegetables, orchards; saves water
- **Sprinkler:** Pipes, sprinkler heads—for larger area; needs pressure

Context:

- **No road:** Tractors and tillers cannot reach. Manual tools, portable solar pumps, or gravity systems dominate.
- **With road:** Power tillers, tractors, and diesel pumps become feasible. Shared or cooperative ownership is common. Farmer-managed irrigation systems account for about two-thirds of Nepal's irrigated area and often outperform agency-managed systems. See References.
- **With electricity:** Electric pump, solar pump—no fuel dependency
- **Maintenance:** Spare parts, repair skill—pump and machinery need upkeep

2.4 Emergency & Backup

- Hand pumps, tube wells
- Community water points
- Emergency tankers during disasters

3 Energy

3.1 Cooking Fuel

- Firewood: primary in remote areas; sustainable harvesting, community forests
- Improved cookstoves such as the *improved chulo*, along with smoke hoods
- Biogas: where livestock exists
- LPG/gas cylinders: where roads allow delivery
- Kerosene: backup; lanterns, stoves
- Solar cookers: experimental
- Electric stoves: where reliable electricity exists
- Agricultural residue: straw, dung cakes

3.2 Electricity

- **Off-grid:** Solar home systems, micro-hydro, and pico-hydro. Rural Nepal has about 3,300 micro-hydro plants. Research ranks micro-hydro as the strongest option for rural electrification, followed by solar. See References.
- Solar lanterns, solar charging stations
- Grid connection that is often unreliable
- Micro-grid, mini-grid
- Backup sources such as diesel generators and batteries
- Street lighting
- Community charging points

3.3 Mechanical & Productive Use

- Grinding mills such as *ghatta*, along with rice hullers
- Irrigation pumps
- Welding, carpentry power tools
- Refrigeration, cold storage
- Human and animal power in places without electricity

3.4 Vehicular & Machine Fuel

- Petrol and diesel where roads exist
- Kerosene for generators
- Electric vehicles and charging as an emerging option

4 Food

4.1 Agriculture

- Seeds include local, improved, and hybrid types, along with seed banks. Research in the mid-hills of Nepal shows that improved cultivars and crop yields are primary drivers of household food self-sufficiency. See Karki et al. 2015 in the References.
- Fertilizers include organic sources such as compost and manure, along with chemical inputs such as urea and DAP.
- Pesticides, herbicides
- Tools include the plough called *halo*, the sickle called *hasiya*, along with *kodalo*, *khurpi*, and the spade.
- Terraces, land leveling
- Agricultural extension, training
- Weather information, crop calendar
- Pest control, crop protection

4.2 Staples

- Rice, wheat, maize, millet, barley, buckwheat
- Staple patterns vary by altitude. The Terai favors rice. Hill areas favor maize and millet. High hills favor buckwheat and potato.
- Pulses, legumes

4.3 Vegetables, Fruits & Other

- Kitchen gardens, seasonal produce

- Fruit trees include citrus, mango, banana, and apple in high-altitude areas
- Spices, herbs
- Storage: drying, pickling, smoking

4.4 Livestock

- Buffalo, cows, goats, sheep, chickens, pigs, ducks
- Milk, meat, eggs, manure, draft power
- Fodder, grazing land
- Veterinary services, vaccination
- Beekeeping and fish farming in ponds

4.5 Storage & Processing

- Grain storage through *bharako* and other forms of rodent protection
- Cold storage in places where electricity exists
- Rice mills, oil mills, flour mills
- Cheese, ghee, and *chyurpi*, which is dried cheese
- Local processing vs. distant

4.6 Markets & Distribution

- Local shops, *haat bazaar*, weekly markets
- Access to district headquarters, wholesale
- Fair price, exploitation
- Food preservation for off-season

4.7 Nutrition & Security

- Dietary diversity
- Fortified foods, iodized salt
- School meals, mid-day meal
- Emergency food, relief
- Community kitchens during disasters
- Malnutrition programs

5 Shelter & Housing

5.1 Construction

- Materials: stone, mud, wood, bamboo, brick, tin roofing
- Earthquake-resistant construction
- Retrofitting of old houses
- Ventilation and chimneys for smoke control
- Insulation for cold areas

5.2 Comfort

- Fireplace through the *chulo* for warmth
- Blankets, bedding
- Lighting: sunlight, fire, kerosene, solar, electric

5.3 Facilities

- Toilet: pit latrine, pour-flush, septic
- Bathing area
- Kitchen, storage

5.4 Resettlement

- Landslide, flood relocation
- Planned resettlement
- Temporary shelter for disaster periods

6 Clothing & Textiles

- Fabric, cloth; local weaving vs. market
- Tailoring, sewing, repair
- Seasonal: winter, monsoon, summer
- Traditional wear: *daura suruwal*, *sari*, *gunyu*
- Footwear: sandals, shoes, boots
- School uniforms
- Washing, drying

Part II

Part II: Transport & Communication

7 Transport

7.1 No Motor Roads

- Foot trails and *ghat* stone steps. Research finds road access is the highest infrastructure priority in rural Nepal, and remote areas show 50–65% lower per capita income. See References.
- Trail maintenance, landslide clearance
- Pack animals such as mules, horses, and yaks in high-altitude regions
- Porters: human carriers
- Ropeways for gravity goods through *tatopani* systems and for passengers
- River transport through boats, rafts, and canoes in the Terai and hill regions
- Bridges: suspension, wooden, rope
- Helicopter: emergency only

7.2 With Motor Roads

- Buses, jeeps, trucks, tractors
- Motorcycles, bicycles
- Ambulance, emergency vehicles
- Goods delivery
- Bus stops, parking

7.3 Infrastructure

- Road construction, maintenance

- Culverts, drainage
- Retaining walls
- Ferry services on rivers
- STOL airstrips in remote areas

8 Communication

8.1 Traditional & Basic

- Postal service, radio through FM and Nepal Radio, word of mouth, messengers, and bulletin boards

8.2 Telephony

- Mobile service through 2G, 3G, and 4G, along with rare landlines and satellite phones for emergencies

8.3 Internet & Digital

- Mobile data, broadband, rare fiber links, cyber cafes, and community internet
- Smartphones, social media, e-governance, online services, digital payments, mobile banking

8.4 Media

- Television, cable, satellite service, and newspapers where available

Part III

Part III: Services

9 Education

9.1 Early Childhood

- Daycare, *bal bikas* centers
- ECD, which means Early Childhood Development
- Mothers' groups, community-run

9.2 School

- Primary levels 1 through 5, secondary levels 6 through 10, and higher secondary levels 11 through 12
- Buildings, classrooms, toilets, drinking water
- Teacher availability, training, housing
- Multi-grade teaching in remote areas
- Textbooks, school supplies, uniforms
- Mid-day meal
- Girls' education, menstrual hygiene
- School management committee

9.3 Higher & Vocational

- Colleges, universities
- Vocational: carpentry, tailoring, agriculture, plumbing, electrician
- Distance learning through internet access
- Scholarships, hostel

9.4 Adult & Non-Formal

- Literacy classes
- Farmer field schools
- Women's groups, cooperatives

9.5 Knowledge & Information

- Library, reading materials
- Bulletin boards, announcements

10 Healthcare

10.1 Primary

- Health posts, sub-health posts
- FCHV, which refers to Female Community Health Volunteers
- Basic medicines, first aid
- Vaccination

10.2 Hospital & Referral

- District, zonal, regional hospitals
- Kathmandu for specialized care
- Ambulance, doko, stretcher, helicopter
- Lab, X-ray, surgery

10.3 Maternal & Child

- Antenatal care and delivery, whether at home or in a facility
- Birthing centers
- Nutrition, immunization

10.4 Specialized

- Dental and eye care, which is often camp-based
- Mental health, which remains underserved
- Disability, rehabilitation
- Elderly, palliative care

10.5 Traditional & Alternative

- *Dhami/jhankri*, Ayurveda
- Herbal medicine

10.6 Pharmacy & Drugs

- Drug availability, quality
- Pharmacy, health post stock

Part IV

Part IV: Governance & Economy

11 Administration & Government

11.1 Offices

- Village or municipal office, described locally as *gaunpalika* or *nagarpalika*
- Ward office, described locally as the *ward karyalaya*
- Ward chairperson, staff
- District office access

11.2 Documentation

- Citizenship, described locally as *nagarikta*
- Birth, death, marriage registration
- Land ownership through the *lalpurja* and related registration
- Tax payment
- PAN, business registration
- Passport, visa

11.3 Justice & Security

- Police post
- Court access
- Mediation, dispute resolution
- Domestic violence, child protection

11.4 Political & Civic

- Election booths, voter registration
- Public hearings, participation
- Citizen assembly, planning

11.5 Programs & Services

- Census, surveys
- Disaster relief
- Employment programs
- Subsidies: fertilizer, seeds, elderly, disabled, single women
- Food security programs

12 Commerce, Finance & Markets

12.1 Markets

- Local shops, *haat bazaar*
- Wholesale, retail
- Weekly, seasonal markets

12.2 Finance

- Banks, branches, mobile banking
- Cooperatives, savings groups
- Microfinance
- Credit, loans
- Remittance transfer

12.3 Insurance

- Life, health, property
- Crop, livestock (emerging)

12.4 Cooperatives

- Agricultural: input supply, marketing
- Consumer
- Credit

Part V

Part V: Environment, Conservation & Sanitation

13 Environment & Natural Resources

- Forests: community, government; fuel, timber
- Grazing land
- Soil conservation, landslide prevention
- Biodiversity, wildlife
- Climate adaptation
- Seasonal calendar: monsoon, drought, cold

14 Resource Conservation

Conserving food, energy, water, and other resources reduces waste, extends availability, and builds resilience—especially where supply is limited or seasonal.

14.1 Food Conservation

- **Cold storage:** Community cold store (electricity or solar-powered) for vegetables, fruits, dairy, meat—extends shelf life, reduces post-harvest loss (20–50% for horticulture without it;

see References), enables off-season supply
- **Evaporative cooling:** *Pot-in-pot*, sand storage—low-tech cooling without electricity
- **Drying:** Sun-drying (*sukeko*), solar dryer—grains, vegetables, fruits, fish, meat (*sukuti*); reduces moisture, prevents spoilage
- **Pickling & fermentation:** *Achar*, *gundruk*, *sinki*, *kinema*—preserves vegetables, extends use
- **Smoking:** Meat, fish—traditional preservation
- **Salt curing:** Fish, meat
- **Grain storage:** *Bharako* (elevated, rodent-proof), neem leaves, ash—protects from pests, moisture
- **Seed storage:** Cool, dry, sealed—maintains viability
- **Root cellar:** Underground or shaded—potato, onion, ginger
- **Reducing waste:** Proper handling, timely harvest, first-in-first-out

14.2 Energy Conservation

- **Cooking:** Improved cookstoves (less fuel), pressure cooker, lid on pot, batch cooking
- **Lighting:** LED bulbs, solar lanterns—efficient use of limited power
- **Insulation:** Houses—reduce heating/cooling need
- **Appliance efficiency:** Energy-efficient fridge, fan, pump—where electricity exists
- **Peak avoidance:** Use mills, pumps during off-peak—if grid with load management
- **Biomass:** Sustainable harvest, replanting—don't deplete forest
- **Waste heat:** Biogas slurry for fertilizer; improved chulo heat for water

14.3 Water Conservation

- **Rainwater harvesting:** Rooftop, tanks—capture monsoon
- **Greywater reuse:** Washing water for garden, livestock
- **Irrigation:** Drip, sprinkler—reduce loss; canal lining
- **Leak repair:** Pipes, taps—prevent waste
- **Spring protection:** Source conservation, recharge
- **Storage:** Tanks, ponds—for dry season

14.4 Material & Other Resource Conservation

- **Reuse:** Containers, bags, tools—repair before replace
- **Recycling:** Metal, plastic—where collection exists; composting organic
- **Timber:** Sustainable harvest, efficient use, alternative (bamboo)
- **Soil:** Compost, mulch, terrace—conserve fertility
- **Medicines:** Proper storage, expiry awareness—avoid waste
- **Fuel:** Efficient stoves, no idle burning

14.5 Community-Level Conservation

- **Grain bank:** Collective storage, buffer for lean season
- **Seed bank:** Preserve local varieties, share
- **Water user group:** Fair allocation, maintenance
- **Forest user group:** Regulated harvest, regeneration

15 Sanitation & Waste

- Toilets: pit, pour-flush, septic; open defecation

- Waste: composting, burning, dumping
- Drainage, flooding
- Solid waste management (rare)

Part VI

Part VI: Social & Livelihood

16 Social, Cultural & Religious

- Temples, *gumbas*, mosques, churches
- Community halls (*samaj ghar*)
- Festivals, *jatras*, ceremonies
- Cemeteries, cremation grounds
- Sports: football, volleyball, cricket, *dandi biyo*
- Youth clubs, women's groups
- Music, dance, theater
- Cinema (where available)

17 Industry, Crafts & Livelihood

17.1 Cottage & Handicraft

- Weaving, knitting
- Pottery, metalwork
- Bamboo, cane work
- Handicraft for market

17.2 Processing & Manufacturing

- Rice mills, oil mills
- Sawmills, brick kilns
- Cheese, ghee, jam
- Small-scale manufacturing

17.3 Livelihood

- Agriculture, livestock
- Wage labor, seasonal migration
- Remittance (major)
- Business, shopkeeping
- Government, NGO employment
- Tourism: homestay, trekking, guide

18 Vulnerable Groups & Social Protection

- Elderly care, pension
- Orphan, child protection
- Disabled support
- Widows, single women

- Dalit, marginalized
- Indigenous, ethnic
- Refugees (if applicable)

19 Technology & Repair

- Mobile repair, battery charging
- Solar installation, repair
- Agricultural machinery repair
- Bicycle, motorcycle repair

Part VII

Part VII: Infrastructure, Security & Governance

20 Public Infrastructure

- Roads, bridges
- Street lighting
- Public spaces, playgrounds
- Warehouses, storage
- Parking, bus stops

21 Security & Emergency

- Natural disasters: earthquake, landslide, flood
- Early warning, evacuation
- Fire: limited services; community response
- Conflict, dispute
- Wild animals, snakes
- Search & rescue
- Relief distribution

22 Legal & Rights

- Marriage, divorce
- Inheritance, property
- Labor rights
- Child rights
- Women's rights
- Land rights, tenancy

23 Identity & Belonging

- Caste, ethnicity
- Language, mother tongue

- Religion
- Voter ID
- Community membership

Part VIII

Part VIII: 100% Fully Independent Village

24 Complete Self-Sufficiency Checklist

What must a village have *within its boundaries* to be 100% fully independent and require no external inputs, services, or markets for survival and functioning? This section catalogs every need for maximum self-reliance. Some items are aspirational. Some have **hard dependencies** that are identified explicitly because they may require minimal external trade or are nearly impossible to produce locally.

24.1 Water Independence

- Own water source: spring, well, river, or sufficient rainwater harvesting
- Own gravity-fed or locally pumped distribution system
- Own purification: boiling, sand filter, solar disinfection—no external chemicals
- Own irrigation: *kulo*, ponds, lift from local stream
- Own maintenance capacity: masons, plumbers
- Water user committee for governance

24.2 Energy Independence

- **Cooking:** Firewood from a sustainably managed community forest, biogas from livestock, and agricultural residue. This excludes LPG and kerosene.
- **Electricity:** Micro-hydro, pico-hydro, or solar—no grid dependency
- **Mechanical:** Water mills (*ghatta*), animal power, human power—or electric mills from local generation
- **Lighting:** Solar, fire, and oil from local seeds such as mustard
- **Hard dependency:** Solar panels, batteries, micro-hydro turbines—typically imported; for true independence would need local manufacture or long-life equipment

24.3 Food Independence

- **Seeds:** Own seed saving, community seed bank—no hybrid dependency
- **Fertilizer:** Compost, manure, green manure, ash—no chemical fertilizer
- **Pesticides:** Neem, local botanicals, cultural methods—no external chemicals
- **Crops:** Full diversity: staples (rice/maize/millet), pulses, vegetables, fruits, spices—sufficient for year-round diet
- **Livestock:** Buffalo/cows (milk, manure, draft), goats, chickens, pigs—for protein, eggs, manure
- **Storage:** *Bharako*, traditional methods—no cold chain
- **Processing:** Own rice mill, oil press, flour mill—powered locally
- **Preservation:** Drying, pickling, smoking, fermentation
- **Hard dependency:** Salt unless there is a local salt source in the Himalayan region. Iodized salt is important for health.

24.4 Shelter Independence

- Own building materials: stone, mud, wood, bamboo, thatch, slate
- Own masons, carpenters, thatchers
- Own repair and construction capacity
- Earthquake-resistant techniques based on local knowledge

24.5 Clothing & Textile Independence

- Own fiber from cotton, wool, hemp, nettle, and silk where sericulture is possible
- Own spinning, weaving
- Own tailoring, repair
- Own dye: natural dyes from plants, minerals
- Own footwear from leather where livestock is available. Rubber soles remain a hard dependency.

24.6 Transport Independence (Internal)

- Foot trails and bridges built locally
- Pack animals: mules, horses, yaks—bred and maintained locally
- Porters using human labor
- No need for motor vehicles if no external travel
- Boats/rafts if river within village

24.7 Education Independence

- Own primary school with teachers from the village who are trained or apprenticed
- Own secondary school when the village has sufficient population to justify it
- Own vocational training through apprenticeship within the village
- Own library and books. Printed books are a hard dependency unless the village relies on oral tradition or handwritten materials.
- Curriculum: state-approved or community-defined
- Own school management, no external funding

24.8 Healthcare Independence

- Own health worker such as a trained FCHV, midwife, or community health worker
- Own pharmacy with herbal medicine, Ayurveda, and basic allopathic supplies. Stocked allopathic medicine remains a hard dependency.
- Own birthing capacity: skilled birth attendant
- Own capacity for: wounds, fractures, common illnesses, dental extraction
- Own mental health support through community systems and traditional healers such as *dhami/jhankri*
- Own vaccination: cold chain impossible without electricity; or accept disease risk
- **Hard dependency:** Antibiotics, vaccines, surgical equipment, complex diagnostics—cannot be produced locally; 100% independence means accepting limits or minimal strategic stockpile from one-time trade

24.9 Governance & Administration Independence

- Own village council, ward committee—elected or customary
- Own dispute resolution through mediation and *panchayat*-style practices
- Own record-keeping: births, deaths, land—written or oral
- Own enforcement: community norms, local guards

- **State interface:** Citizenship, national ID, and land registration remain minimal. The village can maintain internal records. Full independence implies a de facto autonomous unit similar to the historical *rajya* or *thum*.

24.10 Justice & Security Independence

- Own dispute resolution for civil, family, property
- Own security: no external police; community vigilance
- Own disaster response: search, rescue, relief from local stocks
- Own fire response: bucket brigades, community

24.11 Communication Independence

- Internal: word of mouth, village crier, notice boards
- Own postal runner to neighboring villages if any exchange is needed
- Own radio transmitter if electricity and equipment are available. The equipment remains a hard dependency.
- No mobile or internet, unless the village can maintain its own very advanced community network

24.12 Commerce & Finance Independence

- Barter and labor exchange through *parma*
- Own grain bank, seed bank
- Own savings group, rotating fund
- No external currency dependency for internal transactions
- No bank, no remittance—economy is closed

24.13 Sanitation & Waste Independence

- Own toilets: pit, compost—no external septic services
- Own waste: composting, reuse, minimal dumping
- Own drainage: constructed and maintained locally

24.14 Industry, Craft & Repair Independence

- **Blacksmith:** Tools, implements, repair—ore (hard: need local iron or recycling)
- **Carpenter:** Furniture, construction, repair
- **Potter:** Utensils, storage
- **Weaver, tailor:** As above
- **Miller:** Rice, flour, oil
- **Bicycle/motorcycle repair:** Only if vehicles used; parts (hard dependency)
- **Solar/electrical repair:** If solar used; components (hard dependency)

24.15 Raw Materials & Hard Dependencies

- **Salt:** Himalayan rock salt in some areas; otherwise trade
- **Iron/steel:** Mining rare; recycling from scrap; otherwise trade
- **Soap:** Can make from ash + local oil
- **Paper:** Can make from lokta, recycled; or palm leaf, bark
- **Glass:** Cannot produce locally; use oiled paper, open windows
- **Rubber, plastic:** Cannot produce; minimize use
- **Medicines:** Herbal for most; critical allopathic—strategic reserve or accept risk

24.16 Population & Demographics

- Sufficient population to fill all roles: farmers, blacksmith, carpenter, teacher, health worker, etc.
- Marriage pool: internal or arranged with neighboring villages (reduces strict independence)
- Birth rate to sustain population
- Care for elderly, disabled, orphans—within community

24.17 Social, Cultural & Religious

- Own temples, *gumbas*, places of worship
- Own community hall
- Own festivals, rituals, ceremonies
- Own entertainment: music, dance, storytelling
- Own cremation/burial grounds

24.18 Environment & Sustainability

- Community forest: sustainably managed for fuel, timber
- Grazing land for livestock
- Soil conservation, terrace maintenance
- No over-extraction: water, forest, soil

24.19 Time, Calendar & Measurement

- **Timekeeping:** Sundial, water clock, stars, rooster—for work rhythm
- **Calendar:** Lunar (*tithi*), solar—for planting, festivals, rituals
- **Agricultural calendar:** Who knows when to plant? Weather signs, traditional knowledge
- **Measurement:** *Pathi*, *mana*, *dharni* for grain; *haat*, *kosh* for length—standardization within village
- **Value:** How is labor valued in *parma*? How much grain for a day's work? Custom, negotiation

24.20 Surplus, Buffer & Risk

- **Famine reserve:** How many months of grain? One bad year? Two? Strategic storage
- **Winter buffer:** High mountains—6+ months snow; fuel, food, fodder stored
- **Disaster stock:** Earthquake, flood—immediate relief from local stores
- **Crop failure:** Hail, drought, pest—diversity, fallback crops
- **Livestock epidemic:** Quarantine, culling, herbal treatment—no vet

24.21 Geographic & Resource Reality

- **All within boundary:** Arable land, forest, water source, pasture, settlement—must all exist within village territory
- **Altitude:** Hill villages often need land at multiple elevations (rice low, potato high); scattered plots
- **Transhumance:** Seasonal movement of livestock—grazing rights
- **Water source:** Spring, river—must be within or legally secured; upstream villages can cut supply
- **Quarries:** Stone, clay, lime—local deposits or trade

24.22 Skill Redundancy & Succession

- **No single point of failure:** If blacksmith dies, who replaces? Multiple apprentices

- **Knowledge transfer:** Oral, written, apprenticeship—before elder dies
- **Leadership succession:** How is next leader chosen? Hereditary, election, rotation
- **Chicken-egg:** Blacksmith needs anvil; potter needs wheel—initial tools from somewhere; recycling, inheritance

24.23 Death, Ritual & Continuity

- **Cremation/burial:** Wood for pyre, land for burial—who provides?
- **Ritual:** Who performs? Priest, lama—full-time or part-time; how compensated?
- **Inheritance:** Land, house, tools—rules for division; fragmentation over generations
- **Orphan:** Who raises? Kinship, adoption
- **Widow:** Property rights, remarriage—custom

24.24 Conflict, Crime & Punishment

- **Dispute:** Civil, family, property—mediation; what if parties refuse?
- **Crime:** Murder, theft, rape—what punishment? Fine, exile, corporal? No jail without guards, food
- **Factionalism:** Family feud, caste conflict—can split village
- **Boundary dispute:** With neighboring village—grazing, forest, water; no external arbiter

24.25 Defense & External Threat

- **Bandits, raiders:** Historical reality—walls, vigilance, weapons?
- **Wild animals:** Leopard, bear, wild boar—crops, livestock, humans; how to deter?
- **Neighboring village:** Conflict over resources—defense or negotiation

24.26 Healthcare: The Limits

- **Snakebite:** Antivenom—cannot produce; herbal alternatives limited
- **Complicated childbirth:** Breech, hemorrhage—traditional skills can save some; many die
- **Appendicitis, trauma:** Surgery—no surgeon; accept death
- **Cataract, severe dental:** Blindness, chronic pain—camp-based care requires external visit
- **Mental illness:** Severe—restraint, care; no psychiatric drugs
- **Contagious epidemic:** Quarantine, isolation—who enforces?

24.27 Population & Demographics: The Numbers

- **Minimum viable:** Estimate 200–500+ to fill all roles with redundancy
- **Genetic diversity:** Inbreeding if too closed; marriage with neighbors reduces independence
- **Birth/death balance:** Sustain population; family planning—natural methods
- **Disability, chronic illness:** Care within family/community—productivity loss

24.28 Social Obligation & Reciprocity

- **Hospitality:** Cultural duty to feed guests—uses surplus
- **Festival feasts:** Who provides? Rotating, pooling
- **Gift economy:** Marriage, death, festivals—reciprocity, social capital
- **Common labor:** *Shramdan*—trail, canal, temple—who organizes? Who participates?

24.29 Division of Labor: Caste, Gender

- **Traditional:** Blacksmith (*kami*), tailor (*damai*), etc.—caste-based; does independence require breaking or retaining?

- **Gender:** Who fetches water? Who plows? Who weaves? Who cares for children?
- **Child labor:** At what age do children work? School vs. field

24.30 Childcare, Elderly & Disability

- **Childcare:** When parents in field—grandmother, older sibling, communal
- **Elderly:** Bedridden—who feeds, bathes? Family, rotation
- **Disabled:** Born or acquired—who supports? No pension; family, charity

24.31 Storage & Preservation

- **Grain:** Rats, weevils—how to protect? *Bharako* design, neem leaves, smoke—studies report 5–20% cereal storage loss in Nepal; hermetic storage adoption remains low (see References)
- **Seed viability:** Saving year to year—maintain purity, germination

24.32 Pollution & Waste

- **Blacksmith:** Smoke, slag—where?
- **Tannery:** If leather—waste, smell
- **Latrine:** Pit filling—where next? Rotation

24.33 Beauty, Meaning & Spirit

- **Art:** Mural, carving, craft—for meaning, not just function
- **Entertainment:** Music, dance, storytelling—who performs? When?
- **Instruments:** Drum, flute—who makes? Who plays?
- **Why:** Purpose beyond survival—ritual, festival, belonging

24.34 The Hard Truths

1. **100% is impossible** without accepting death, disability, and suffering that modern medicine could prevent.
2. **Geographic luck** matters: a village without local iron, salt, or sufficient water cannot be independent.
3. **Population** must be large enough for all skills plus redundancy—small villages cannot.
4. **Marriage** with neighbors breaks strict independence; inbreeding breaks health.
5. **Surplus** is essential—for bad years, festivals, guests—but surplus requires good land, labor, and luck.
6. **Conflict** without external arbiter can destroy a village from within.
7. **Independence** was historically possible when villages were *rajya*, *thum*—semi-autonomous. The modern state, market, and aspiration have made it rare.

24.35 Summary: Independence vs. Reality

A 100% fully independent village is theoretically possible only under four conditions. It must produce or harvest everything from local resources. It must have sufficient population and skill diversity. It must either accept **hard dependencies** for a few items such as salt, iron, and critical medicines through minimal trade, or accept higher risk from disease and crop failure without an external buffer. In practice, *maximum self-reliance* is the achievable goal because it minimizes external dependency while maintaining dignity and resilience.

25 Summary: The Continuum

Fully Remote Local water, firewood, subsistence agriculture, foot/mule transport, health posts, primary schools, ward office, radio, occasional mobile. No grid, no roads, no internet.

Partially Connected
Micro-hydro or solar, trail/ropeway, secondary school access, district hospital referral, weekly market, patchy mobile.

Fully Connected
Roads, grid, internet, banks, higher education, better healthcare, LPG, vehicles, e-services.

100% Independent
Maximum self-reliance: local water, fuel, food, shelter, education, healthcare, governance, industry—minimal external trade for salt, iron, critical medicines.

Understanding this continuum helps in planning interventions, policy, and development projects tailored to each village's reality.

Part IX

Part IX: Village Needs Tree — Technical Specification

26 Overview

The Village Needs Tree is a YAML tree with metadata. **No GUI.** The metadata encodes all dependencies between nodes (families/leaves and their ancestors).

26.1 Files

File	Purpose
`village_governance_tree.yaml`	Canonical governance model authoring file in normalized node-registry form, with explicit parent/child links and per-node dependency metadata.
`village_needs_catalog.yaml`	Canonical village needs catalog (id, scope, category, barriers, feasibility, evidence).
`budget_scenarios.yaml`	Named budget scenarios for intervention planning.
`dist/village_needs_catalog.json`	Generated JSON export of the canonical needs model.
`dist/village_governance_public_data.json`	Full public data bundle with model and analysis outputs.
`dist/village_governance_dashboard_data.json`	Compact frontend-ready bundle for the public dashboard.
`village_needs_schema.json`	JSON schema for the village needs model.
`validate_village_model.py`	Validates `village_needs_catalog.yaml` and tree semantics.
`check_needs_coverage.py`	Verifies all needs are covered in the tree.
`extract_dependency_edges.py`	Extracts dependency graph as edges.
`generate_governance_diagrams.py`	Generates ASCII, Mermaid, and Graphviz diagrams.
`analyze_dependency_routes.py`	Route-length analysis, self-sufficiency score, push-down opportunities, and budget scenarios.
`export_public_data.py`	Generates public JSON bundles from canonical YAML.
`public/`	Static public dashboard consuming generated frontend-ready JSON.

26.2 Meta = Dependencies

Each node in the tree has a `meta` block:

satisfies: Needs met within this node (no external dependency).
requires: `{node, need}` — this node depends on that node for that need. Creates dependency edges.
provides: Needs this node supplies to children/others.
need_deps: Need-to-need dependencies within this node.

The `requires` field creates the dependency graph: household → ward (water), household → village (market, school).

26.3 Governance Node Model

```
root_node: village
nodes:
    - id: village
        type: V
        parent_id: null
        child_ids: [ward_1, ward_2]
```

```
  - id: ward_1
      type: W
      parent_id: village
      child_ids: [n1_1, n1_2]

  - id: n1_1
      type: N
      parent_id: ward_1
      child_ids: [h1_1_1, h1_1_2, h1_1_3]

  - id: h1_1_1
      type: H
      parent_id: n1_1
      child_ids: []
```

Internally, the implementation now normalizes the governance model into a generic node registry rather than depending directly on a hardcoded nested YAML shape. That makes the runtime model more modular and less repetitive. The same representation can support the current minimal village, future additional layers, and real structural changes such as household split, household merge, neighbor-group reorganization, and ward boundary change.

Optional lineage support. A node may include lineage metadata such as `split_from`, `merged_from`, or `merged_into`. These fields are optional in the minimal prototype, but the framework now reserves space for them so real institutional change can be represented without changing the code architecture.

26.4 Completeness Verification

```
python check_needs_coverage.py
```

Verifies all 65 needs (minus external) are satisfied or required somewhere in the governance model.

26.5 Diagram Generation

```
python generate_governance_diagrams.py
```

Produces:

- `village_governance_tree_ascii.txt` — ASCII tree
- `village_governance_tree_mermaid.mmd` — Mermaid tree (view in VS Code, GitHub)
- `village_dependency_routes_mermaid.mmd` — Mermaid dependency graph
- `village_governance_tree_graphviz.dot` — Graphviz DOT
- `village_governance_tree_graphviz.png` — PNG (if `dot` installed)

26.6 Extending the Governance Model

1. **Add nodes:** Extend `village_governance_tree.yaml` with new governance nodes and connect them through `parent_id` and `child_ids`.
2. **Add meta:** For each node, set satisfies/requires/provides.
3. **Record change history:** When a household splits or multiple households merge, preserve identity through optional lineage fields rather than deleting the historical relationship.
4. **Add needs:** Extend `village_needs_catalog.yaml`; reference in meta.

27 Taxonomy and Classification System

27.1 Scope Levels (Production-Consumption Scale)

Level	Code	Definition	Size	Example
Household	H	Single family unit.	1 family	Own rice, own chulo, own toilet
Neighbor	N	2–5 adjacent households.	2–5 families	Parma, borrowing tools, shared well
Ward	W	Ward or neighborhood.	10–50 families	Ward trail, ward water point, local mill
Village	V	Entire village.	100–500+ families	Haat bazaar, health post, school

Rule: A need is satisfied at level X if the production-consumption cycle is contained within that level.

27.2 Scope of a Need

Each need has:

min_scope: Minimum level at which this need can be satisfied (H, N, W, or V).

scope_options: Levels at which this need CAN appear (a need may be satisfiable at multiple levels).

Examples:

- **Drinking water:** scope_options = [H, N, W, V]. H: Own well, rainwater. N: Shared well with 2–3 neighbors. W: Ward tap. V: Village piped system.
- **Haat bazaar:** scope_options = [V] only (cannot exist at H, N, W).
- **Own rice consumption:** scope_options = [H] (shortest cycle).

27.3 Cycle Length

Length	Code	Definition
Short	S	Production → consumption within same unit; minimal intermediaries
Medium	M	2–3 steps; involves neighbors or ward
Long	L	Many steps; involves village institutions, markets, money, storage

27.4 Need Attributes (Extensible Schema)

Attribute	Type	Required	Description
id	string	yes	Unique identifier (e.g., `water.drinking.spring`)
name	string	yes	Human-readable name
category	string	yes	water, energy, food, shelter, etc.
subcategory	string	no	Finer grouping
scope_options	[H,N,W,V]	yes	Levels at which this need can be satisfied
min_scope	H|N|W|V	yes	Minimum scope (derived from scope_options)
cycle_length	S|M|L	no	Production-consumption cycle length
depends_on	[id]	no	Other needs this one requires
barriers	[barrier]	no	Why this need stays at higher scope than minimum (see below)
feasibility	object	no	Cost, maintenance, difficulty, and next-scope planning data
confidence	enum	no	Confidence in the claim: low | medium | high | provisional
evidence	[object]	no	Citations, observations, or notes supporting the claim
notes	string	no	Context, Nepal-specific
source_section	string	no	Reference to section in this document

27.4.1 Barrier Objects

Each barrier documents *why* a need is not yet at its lowest possible scope. This is the key to extensibility: the system separates **what is physically possible** (`scope_options`) from **what currently prevents it** (`barriers`).

Field	Type	Required	Description
type	string	yes	cost | expertise | population | infrastructure | geography | regulation
description	string	yes	Why this barrier exists
holds_at	H|N|W|V	no	Scope below which this barrier prevents push-down
removable	boolean	no	Can this barrier be removed with investment?

27.4.2 Feasibility Objects

Barriers explain *why* a need remains above its minimum possible scope. Feasibility explains *how hard it would be* to push it down in practice.

Field	Type	Req.	Description
target_scope	H|N|W|V	no	Recommended next target scope
investment_cost_band	1–5	no	1=low capital, 5=very high capital
maintenance_cost_band	1–5	no	1=low maintenance, 5=very high maintenance
`implementation_difficulty_band`	1–5	no	1=easy, 5=very difficult
time_horizon	enum	no	immediate | near_term | medium_term | long_term
notes	string	no	Planning assumptions and local context

These are ordinal planning bands, not exact price claims. Their role is to make interventions comparable before a detailed budget exists.

27.4.3 Evidence and Confidence

The model now distinguishes between *what is claimed* and *how strongly that claim is supported.*

Field	Type	Required	Description
confidence	enum	no	low | medium | high | provisional confidence in the modeled claim
evidence.kind	string	no	citation | observation | planning_judgment
evidence.citation	string	no	Reference title or source label
evidence.url	string	no	Link to source when available
evidence.notes	string	no	Short statement of what the evidence supports

This makes the model auditable. A public user can now ask not only "why is this need placed here?" but also "how certain is that claim, and what supports it?"

Barrier types:

cost: Equipment, materials, or ongoing expense too high for current wealth. *Removable:* when resources increase.

expertise: Skilled labor or training required but not available. *Removable:* with education/vocational programs.

population: Need requires minimum number of people (e.g., school needs enough children). *Typically structural.*

infrastructure: Depends on other infrastructure (electricity, roads) not yet present. *Removable:* with investment.

geography: Physical resource is shared or location-specific (forest, spring). *Often structural.*

regulation: Legal or regulatory requirement mandates collective ownership. *Often structural.*

Design principle: When a removable barrier is overcome (e.g. the village becomes rich enough to buy each family a hand mill), the only change needed is:

1. Move the need from the higher-scope node's `satisfies` to the lower-scope node's `satisfies` in `village_governance_tree.yaml`.
2. Update the household's `requires` to remove that dependency.
3. Run `python analyze_dependency_routes.py` to verify the score improved.

No code changes. Only model edits.

27.5 Self-Sufficiency Analysis

The system provides four analysis modes via `analyze_dependency_routes.py`:

Default report:
Route lengths, per-household scores, push-down opportunities, barriers, irreducible dependencies, and self-sufficiency score.

`--whatif <need> <scope>`:
Simulate pushing a need to a lower scope and compare scores. Shows current vs. projected score and lists barriers to overcome.

`--barriers`: List all barriers with removable/structural classification. Removable barriers are the investment targets.

`--feasibility`:
Rank interventions by total route savings against investment, maintenance, and implementation difficulty bands.

`--budget <band>`:
Select the best intervention portfolio within a given investment budget band.

`--scenarios`: Run named budget scenarios from `budget_scenarios.yaml` and print their selected interventions.

Example: `python analyze_dependency_routes.py --whatif food.processing.mill H` answers "What if every household had its own mill?" and reports the score improvement and cost/infrastructure barriers to overcome.

The feasibility ranking uses two derived indicators:

- **ROI:** total hop savings divided by investment cost band.
- **Priority:** total hop savings divided by the sum of investment, maintenance, and implementation difficulty bands.

This does not replace detailed budgeting. It creates a rigorous first-pass ranking of which interventions deserve deeper planning first.

The current analysis implementation uses the actual containment hierarchy to measure governance-route length. That means the route engine no longer depends on a hardcoded assumption that the only valid path is H → N → W → V. As long as the taxonomy defines the layers and the governance model connects nodes consistently, the analysis remains valid.

Named scenarios convert that ranking into concrete bundles such as bootstrap, practical, and transformation paths. This gives the system a direct bridge from structural diagnosis to phased decision making.

27.6 Extensibility

- **Add new need:** Add entry to `village_needs_catalog.yaml`; assign id, scope_options, category. Optionally add barriers, feasibility, confidence, and evidence.
- **Add feasibility:** If a push-down is realistic, add a `feasibility` block with cost, maintenance, and difficulty bands.
- **Add evidence:** Attach citation, local observation, or planning judgment to make the claim reviewable.
- **Push a need down:** Edit `village_governance_tree.yaml` — move need from higher node's satisfies to lower node's satisfies; update requires. Run analysis.
- **Add new barrier type:** Add to `village_needs_schema.json` barrier type enum; add to relevant needs.
- **Add new level:** Extend taxonomy scope levels with a new code and order; the traversal, validation, and analysis layers now derive hierarchy rules from taxonomy rather than from

hardcoded loops.

- **Add new attribute:** Add to schema; existing needs get null/default.
- **Add new village:** New governance-model instance; same need set, different node population.
- **Future wealth scenario:** Use `--whatif` to preview; then edit YAML when ready. No code changes needed.

27.7 Completeness Check Procedure

For each need in this document:

1. Does it have an id?
2. Is scope_options assigned?
3. Is category assigned?
4. Are dependencies (if any) listed?

27.8 Version

- Schema version: 1.0
- Taxonomy version: 1.0

28 How to Classify a Need

28.1 Step 1: Identify the Need

From this document, extract the need. Give it a unique `id`:

- Format: `category.subcategory.item`
- Examples: `water.drinking.spring`, `food.market.haat`, `energy.cooking.biogas`

28.2 Step 2: Assign scope_options

Question: At which levels can this need be satisfied?

If the need...	scope_options
Can be done by 1 family alone	`["H"]`
Requires 2–5 families (neighbors)	`["N"]` or `["H","N"]` if both possible
Requires ward (10–50 families)	`["W"]` or include in array
Requires village (100+)	`["V"]` or include in array

Examples:

- **Own rice consumption:** Family grows, family eats. → `["H"]`
- **Drinking water:** Own well (H), shared well (N), ward tap (W), village pipe (V). → `["H", "N", "W", "V"]`
- **Haat bazaar:** Only exists at village scale. → `["V"]`
- **Parma:** By definition needs 2+ families. → `["N"]`
- **Health post:** Serves many. → `["W","V"]`

28.3 Step 3: Assign cycle_length

Length	When
S (short)	Production → consumption within same unit; minimal steps
M (medium)	2–3 steps; involves neighbors or ward
L (long)	Many steps; village institutions, markets, money, storage

28.4 Step 4: List depends_on (if any)

If this need requires another need, add its id:

- Cold storage depends on electricity → `"depends_on": ["energy.electricity.solar"]`
- Pump irrigation depends on pipes → `"depends_on": ["water.irrigation.pipes"]`

28.5 Step 5: Add to village_needs_catalog.yaml

```
- id: category.subcategory.item
    name: Human-readable name
    category: water|energy|food|shelter|...
    subcategory: optional
    scope_options: [H, N, W, V]
    cycle_length: S|M|L
    depends_on: []
    barriers: []
    feasibility: {}
    notes: Context if needed
    source_section: Water.Drinking
```

28.6 Decision Tree: scope_options

```
Can 1 family do it alone?
  YES --> Include "H"
  NO  --> Don't include "H"

Can 2-5 neighbors do it together?
  YES --> Include "N"
  NO  --> Don't include "N"

Does it need ward-scale (10-50)?
  YES --> Include "W"
  NO  --> Don't include "W"

Does it need village-scale (100+)?
  YES --> Include "V"
  NO  --> Don't include "V"
```

28.7 Ambiguous Cases

- **Mobile phone:** Handset = H; tower = V. Classify "Mobile connectivity" → V (infrastructure); "Using mobile" → H (consumption).
- **Mill:** Hand mill = H; electric mill = W or V. Use scope_options: `["H","N","W","V"]` with note.
- **School:** Building = V; child attending = H (but depends on V). Classify "School" as V.

29 Metadata Schema for Tree Nodes

The metadata on each node encodes **dependencies between nodes**. This is what creates the dependency graph—no separate GUI needed.

29.1 Core Meta Fields

Field	Type	Description
`satisfies`	[need_id]	Needs satisfied *within* this node
`requires`	[{node, need}]	Needs from *other* nodes. Creates dependency edge.
`provides`	[need_id]	Needs this node *supplies* to others
`need_deps`	[{need, depends_on}]	Need-to-need dependencies within this node

29.2 Dependency Semantics

satisfies: "We do it ourselves." No external node needed.
requires: "We get it from them." Example: `{node:"ward_1",need:"water.drinking.piped"}` = this node depends on ward_1 for water.
provides: "We supply this to others." Parent/ancestor provides to children.
need_deps: "Need A needs need B." Both at same node; internal dependency.

29.3 Example: Household

```
meta:
  satisfies:
    - food.staples.own      # own rice, own consumption
    - energy.cooking.firewood
    - shelter.toilet
  requires:
    - {node: ward_1, need: water.drinking.piped}
    - {node: village, need: food.market.haat}
    - {node: village, need: education.school}
```

Interpretation: This household is self-sufficient for staples, cooking, toilet. It depends on ward_1 for water, village for market and school.

29.4 Example: Ward

```
meta:
  provides:
    - water.drinking.piped
  requires:
    - {node: village, need: water.drinking.committee}
  satisfies:
    - transport.trail   # ward maintains its trail
```

29.5 Extensibility

Add fields as needed:

- `cycle_length`: S|M|L for this node's aggregate cycle
- `scope`: H|N|W|V
- `notes`: Free text

30 Recommended Architecture for a Public System

The current repository already has a strong domain idea, but its main weakness is that one conceptual system is spread across several machine formats and utility scripts. For a **minimal**

but complete public governance system, the design should move toward **one canonical source of truth, many generated views**.

30.1 What Should Stay

- **This document:** the human-readable constitution, design philosophy, and formal explanation of the system.
- **Machine-readable model:** the canonical governance data.
- **Generated artifacts:** diagrams, reports, public summaries, and scenario outputs.

30.2 What Should Be Reduced

The repository should avoid multiple hand-maintained representations of the same truth.

- Do not hand-edit both YAML and JSON for the same fact.
- Do not treat generated diagrams as authoritative data.
- Do not encode policy logic separately in many scripts.

30.3 Target Structure

The recommended target structure is:

1. **Canonical model:** one source of truth for needs, barriers, deployment, and routing.
2. **Validation layer:** schema validation and semantic rules.
3. **Analysis layer:** route scoring, push-down analysis, scenarios, and cost tradeoffs.
4. **Presentation layer:** PDF, public website, interactive graph, searchable tables, and proposal forms.

30.4 Canonical Data Model

The cleanest long-term model separates four things:

Need definition:
What the need is, its scope options, dependencies, and barriers.

Deployment: Where the need is actually satisfied today in a specific village.

Scenario: What would happen if a barrier were removed or a need were pushed down.

Evidence: Why the model makes that claim: references, cost assumptions, local observations.

This separation is important. A need may be *possible* at H-scope, *currently deployed* at V-scope, and *blocked* by cost. Those are different facts and should remain different fields.

30.5 Recommended Technology Direction

There are three reasonable maturity levels.

30.5.1 Level 1: Minimal Repository System

Use one canonical authoring file format and generate everything else.

- Preferred authoring format: **YAML** for human editing.
- Preferred validation/runtime layer: **Python** with schema validation and analysis.
- Public outputs: PDF, JSON export, generated diagrams, CSV tables.

This is the best fit if the project remains primarily research/design oriented.

30.5.2 Level 2: Public Open Data System

Keep the canonical model in a repository, but generate a public site from it.

- Static site generator: Astro or Next.js.
- Graph visualization: Cytoscape.js or D3 instead of static Mermaid only.
- Public data exports: JSON, CSV, and printable summaries.
- Public modification method: proposal-based edits with review, not direct raw file editing.

This is the best fit if the public should be able to explore the model but not directly mutate the source of truth.

30.5.3 Level 3: Full Governance Platform

If many villages, planners, or agencies will collaborate, move from repo-centered editing to an application.

- Backend: FastAPI, Django, or a typed TypeScript backend.
- Database: PostgreSQL or SQLite first, Postgres when multi-user collaboration is needed.
- Editing UI: forms for needs, barriers, village deployment, and scenarios.
- Audit trail: every change recorded as proposal, approval, and rationale.

At this stage the repo becomes the export surface, not the primary editing surface.

30.6 Presentation to the Public

To make the system understandable to non-technical users, the public interface should not expose raw YAML/JSON first. It should expose questions and answers.

1. What does a village need?
2. Which needs are local and which require higher coordination?
3. What makes this village dependent?
4. Which barriers are removable with investment?
5. What score improvement happens if one barrier is removed?

That implies a public site with the following views:

- **Need catalog:** searchable list of all needs with scope, barriers, and references.
- **Village map/tree view:** households, neighbor groups, wards, village institutions.
- **Dependency graph:** interactive routes from household to source of need satisfaction.
- **Scenario simulator:** "What if every ward had a school?" or "What if households had hand mills?"
- **Barrier dashboard:** removable vs. structural barriers.
- **Policy summary:** top interventions that reduce route length most efficiently.

The current repository now implements the first public step of this design: `export_public_data.py` generates `dist/village_governance_dashboard_data.json`, and the `public/` directory contains a static dashboard that renders summary cards, top interventions, budget scenarios, a searchable need catalog, and a filtered dependency graph.

30.7 Modification Model

For safe public modification, raw file editing is the wrong surface. The correct surface is:

1. User proposes a change in a form.
2. System validates it against schema and semantic rules.
3. System previews route/score impact.

4. Reviewer approves or rejects with rationale.
5. Canonical model updates and exports regenerate.

This makes the system both public-facing and governable.

30.8 Implemented Immediate Structural Improvements

The current repository now includes the following high-value structural improvements:

1. One canonical human-edited machine source of truth for needs in `village_needs_catalog.yaml`, with generated JSON exports.
2. Semantic validation beyond schema: target node must actually satisfy/provide the required need; tree scope must respect allowed scope options; evidence and confidence must be structurally valid.
3. Explicit feasibility and budget-scenario planning so route minimization can be compared against cost and implementation effort.
4. A public export step that produces a compact, frontend-ready dashboard bundle.
5. A static public dashboard as the initial public interface, ready to be extended into richer interactive views.

30.9 Minimality Principle

Minimality here should not mean fewer ideas. It should mean fewer *authoritative representations*. The ideal state is:

> One canonical model, one main human document, one command surface, and many generated outputs.

That is the most maintainable way to remain 100% comprehensive while still being easy to explain, modify, and publish.

Part X

Part X: Decision and Feasibility Review

31 One Hundred Questions Before Committing to Format, Tooling, and Public Platform

Strong governance design should not jump from preference to implementation. The following ten groups of ten questions form a decision gate before major architectural choices such as "YAML vs. JSON", "repo vs. database", or "static site vs. application".

31.1 A. Mission and Scope

1. What is the precise primary objective: documentation, planning, simulation, coordination, or public governance?
2. Is the system for one village, many villages, or a national template?
3. What must remain minimal, and what must remain comprehensive?
4. Which parts are normative policy, and which parts are descriptive observation?
5. Is the unit of governance the household, the ward, the village, or multiple nested levels equally?
6. What decisions must the system support in the next 12 months?
7. What decisions might the system support in the next 10 years?

8. Which users matter most at launch: researchers, planners, villagers, NGOs, or the public?
9. What would count as failure even if the data model is technically correct?
10. What would count as success even if the system remains technologically simple?

31.2 B. Canonical Data Model

1. What is the single canonical source of truth?
2. Which facts belong in the need definition versus the village deployment?
3. Which facts are timeless taxonomy versus context-specific local data?
4. Should barriers belong to needs, deployments, or both?
5. How should evidence, citations, and local observations be attached to claims?
6. How should uncertainty be represented: confidence, unknown, disputed, provisional?
7. How should external dependencies be encoded distinctly from internal dependencies?
8. Should costs be stored as absolute numbers, ranges, or scenario assumptions?
9. How should time be represented: current state, target state, planned state, historical state?
10. How will the model evolve without forcing incompatible rewrites?

31.3 C. YAML vs. JSON Decision

1. Is the primary author a human editor or a machine pipeline?
2. Will comments and policy notes be important in the canonical data?
3. Does the team need a format that non-programmers can edit directly?
4. Are strict machine contracts more important than human readability?
5. Can the team enforce one serializer and one formatting standard?
6. Will multiple tools consume the model through generated JSON anyway?
7. Is it acceptable to keep JSON only as an export rather than a source?
8. How will accidental YAML ambiguity be prevented?
9. How will field order be stabilized for reviews and diffs?
10. Is the chosen canonical format reversible if the platform later moves to a database?

31.4 D. Governance Semantics

1. What exactly does “satisfies” mean operationally?
2. What exactly does “provides” mean operationally?
3. Can a node provide something it does not satisfy internally?
4. When is a dependency considered acceptable rather than a design weakness?
5. How should collective ownership be modeled differently from service access?
6. How should legal requirements be distinguished from practical constraints?
7. How should seasonal availability be represented?
8. How should emergency-only pathways be represented separately from normal pathways?
9. How should quality differences be represented, not just binary availability?
10. What is the formal meaning of self-sufficiency in this system?

31.5 E. Validation and Data Quality

1. Which errors are structural, which are semantic, and which are policy judgments?
2. Should validation fail if a target node does not actually provide the required need?
3. Should validation fail if a node satisfies a need at a disallowed scope?
4. Should barriers be mandatory whenever a need is deployed above minimum scope?
5. How should contradictions between citations and local practice be handled?
6. Should incomplete entries be allowed in draft mode?
7. How should deprecated needs or old categories be handled?
8. How will the system catch drift between technical documentation and machine data?

9. What validations belong in the command line versus the public UI?
10. Which validations must block publication?

31.6 F. Feasibility and Economics

1. What is the cost to push each removable barrier down one level?
2. What is the maintenance burden after the initial investment?
3. Which interventions save the most route length per unit cost?
4. Which interventions create fragility even while shortening routes?
5. When is shared infrastructure economically superior to household independence?
6. How should labor availability be modeled as a real constraint?
7. How should training time be modeled as a cost?
8. Which technologies are repairable locally and which create new dependency?
9. How should replacement cycles be modeled for equipment like solar or pumps?
10. Which "self-sufficiency" choices are symbolic but economically weak?

31.7 G. Public Presentation

1. What should a villager understand in under three minutes?
2. What should a planner understand in under ten minutes?
3. What should a researcher be able to inspect in raw form?
4. Which public views are essential on day one?
5. Should the public see raw routes or summarized indicators first?
6. How should complex dependency graphs be made understandable on mobile devices?
7. How should Nepali and English coexist in the public interface?
8. Which outputs should be printable for low-connectivity environments?
9. What should be interactive versus static?
10. Which public outputs must be generated automatically from the canonical model?

31.8 H. Editing and Decision Workflow

1. Who is allowed to propose a change?
2. Who is allowed to approve a change?
3. What evidence is required before adding a new need or barrier?
4. Should every change include a rationale and expected route impact?
5. Should the system preview score changes before approval?
6. How should disputes over taxonomy be resolved?
7. How will local knowledge be incorporated without losing consistency?
8. How should experimental scenarios be separated from approved policy state?
9. How will major version changes be governed?
10. What is the rollback procedure if a bad change is approved?

31.9 I. Platform Growth

1. When does a repo stop being enough?
2. At what scale does a database become necessary?
3. When does a static site stop being enough?
4. Which APIs will external consumers eventually need?
5. How should authentication be introduced if public editing is added later?
6. Which parts of the system should remain exportable as plain files forever?
7. How should village-specific customization coexist with a shared national taxonomy?
8. How should scenario engines evolve without breaking the canonical model?
9. What must remain offline-capable?

10. Which technical choices today would make later scaling unnecessarily hard?

31.10 J. Decision Gate

1. Is the current decision being made because it is truly better, or only because it is familiar?
2. What assumptions are still untested?
3. What would change the decision if new evidence appeared next month?
4. What is the cost of choosing too early?
5. What is the cost of delaying too long?
6. Which decision is reversible and which is hard to reverse?
7. Can the next version preserve optionality while still delivering value now?
8. Which smallest improvement would reduce confusion immediately?
9. If the system doubled in complexity, would the chosen architecture still be intelligible?
10. If the public had to trust this system, what proof of rigor would they require?

32 Current Decision

Based on the present repository and the questions above, the strongest immediate decision is:

1. Keep `village_needs_schema.json` as JSON.
2. Make `village_needs_catalog.yaml` the canonical human-edited source.
3. Keep `village_governance_tree.yaml` as canonical deployment structure.
4. Keep `budget_scenarios.yaml` as canonical named planning input for budget choices.
5. Generate JSON only as an export artifact for tools and public interfaces, including the dashboard bundle.
6. Delay any move to a database or full application until multi-user editing and approval workflows become real needs.

This preserves human readability, keeps the system extensible, and avoids premature platform complexity.

Part XI

Part XI: References

33 References and Further Reading

The following research articles, books, and reports inform this framework. Links and citations are provided for deeper study.

33.1 Infrastructure & Wellbeing

- **Access to infrastructure and human well-being: evidence from rural Nepal.** Examines how roads, water, irrigation, electricity, and other infrastructure affect wellbeing; finds roads highest priority, followed by drinking water and irrigation; remote areas show 50–65% lower per capita income, higher malnutrition and dropout. `https://scispace.com/pdf/access-to-infrastructure-and-human-well-being-evidence-from-4uuy9g9dkt.pdf`
- **Rural Living in Nepal: Housing, Water & Waste in Paanchkhal.** NPRC Journal of Multidisciplinary Research. Housing (Pakka/Kachcha), water sources, purification, latrines, waste disposal. `https://nepjol.info/index.php/nprcjmr/article/view/78324`

33.2 Water & Irrigation

- **Does Rural Water System Design Matter? A Study of Productive Use of Water in Rural Nepal.** Water 11(10), 1978. Productive use of water beyond drinking. https://www.mdpi.com/2073-4441/11/10/1978
- **Functionality of Rural Community Water Supply Systems and Collective Action: Guras Rural Municipality, Karnali.** https://nppr.org.np/index.php/journal/article/view/26/53
- **Sulitar Irrigation System: Struggle from Poverty to Prosperity.** Hydro Nepal. Farmer-managed irrigation, participatory approach, community ownership. https://www.nepjol.info/index.php/HN/article/view/13276
- **Farmers in Nepal reap benefits of rehabilitated farmers-managed irrigation systems.** Asian Development Bank. FMIS rehabilitation, crop intensification, income gains. https://www.adb.org/results/farmers-nepal-reap-benefits-rehabilitated-farmers-managed-irrigation-systems
- **Improving irrigation governance and management in Nepal.** Virginia Tech. 20+ years of research on community resource management. https://vtechworks.lib.vt.edu/items/cbc1aa78-7c13-4404-bfbf-2552edcc72e7
- **Irrigation in Nepal: Opportunities and Constraints.** Third World Centre. https://thirdworldcentre.org/wp-content/uploads/2020/07/RPP-Dec-1-1989-Irrigation-in-Nepal-Opportunities-and-Constraints.pdf

33.3 Food Security & Post-Harvest

- **Karki TB, Sah SK, Thapa RB, McDonald AJ, Davis AS (2015) Identifying Pathways for Improving Household Food Self-Sufficiency Outcomes in the Hills of Nepal.** PLoS ONE 10(6): e0127513. CART analysis; agronomic management, maize-fingermillet yields, improved cultivars as primary drivers. https://doi.org/10.1371/journal.pone.0127513
- **Post-harvest practices of horticultural crops in Nepal: Issues and management.** 20–50% post-harvest loss; cold storage, packaging. https://www.academia.edu/66093653/Post_harvest_practices_of_horticultural_crops_in_Nepal_Issues_and_management
- **On-Farm Grain Storage and Challenges in Bagmati Province, Nepal.** Sustainability. 5–20% storage loss; pests, monsoon; hermetic storage adoption low. https://www.cgiar.org/research/publication/on-farm-grain-storage-and-challenges-in-bagmati-province-nepal/
- **Assessing the Impact of Hermetic Storage Technology on Post-harvest Losses among Maize Farmers in Nepal.** Preprints. https://www.preprints.org/manuscript/202411.1421/v1

33.4 Energy

- **Techno-Economic Modelling of Micro-Hydropower Mini-Grids in Nepal.** Energies. Financial sustainability, electric cooking, productive use. https://www.mdpi.com/2071-1050/14/16/4232
- **Identifying the best decentralized renewable energy system for rural electrification in Nepal.** Journal of Asian Rural Studies. Micro-hydro ranked best, then solar. http://pasca.unhas.ac.id/ojs/index.php/jars/article/view/2097
- **Performance evaluation of solar PV mini grid system in Nepal: Thabang and Sugarkhal.** Frontiers in Energy Research. https://www.frontiersin.org/journals/energy-research/articles/10.3389/fenrg.2023.1321945/full
- **Sustainability assessment of a micro hydropower plant in Nepal.** Energy, Sustainability and Society. https://energsustainsoc.biomedcentral.com/articles/10.1186/s

13705-018-0147-2

33.5 Health & Education

- **Perception of Health and Care Seeking Behaviors in High-Altitude Villages of Rural Nepal.** HPROSPECT. Traditional healers, cultural beliefs, accessibility. https://nepjol.info/index.php/HPROSPECT/article/view/24103
- **Assessing the health impacts of a Comprehensive Rural Health Project in Nepal.** PLOS Global Public Health. Village Alive Project, Dalit villages. https://journals.plos.org/globalpublichealth/article?id=10.1371/journal.pgph.0004458

33.6 Rural Development & Transport

- **Rural Transportation Infrastructure in Low- and Middle-Income Countries.** Sustainability. Impacts, interventions, links to agriculture and health. https://www.mdpi.com/2071-1050/14/4/2149
- **Handbook of Rural Development.** Comprehensive rural development scope.
- **Infrastructure for Smart Villages.** Routledge. Fit-for-purpose rural infrastructure, SDG alignment.
- **UN Handbook on Infrastructure Asset Management.** Managing infrastructure for sustainable development.